



Tripartite Game of Industry-Education Integration from the Perspective of Interest Evolution

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Abstract: This study focused on three main entities involved in industry-education integration: Universities, enterprises, and the government. Based on the evolutionary game theory, a tripartite evolutionary game model was constructed to thoroughly evaluate the dynamics between interest evolutionary game relationships and strategic choices among these entities during the process of industry-education integration. By setting six strategies such as “active cooperation” and “passive cooperation” for universities, “deep participation” and “formal participation” for enterprises, and “regular supervision” and “passive supervision” for the government, the study systematically analyzed the stability conditions for each entity’s strategic choices through a payoff matrix and replication dynamic equations. It was found that the stability of strategic choices for universities, enterprises, and the government depended on the comparison of net benefits under different strategies. When the net benefits of active cooperation, deep participation, and regular supervision exceeded those of their respective passive strategies, each entity tended to choose the active strategy owing to its stability. Otherwise, they would lean towards passive strategies. This study revealed the inherent laws governing the strategic choices of the three entities in industry-education integration, thus providing a theoretical basis and policy recommendations for optimizing integration policies and promoting tripartite cooperation. It holds significant importance for driving high-quality development in the proposed industry-education integration.

Keywords: Interest evolution; Industry-Education integration; Tripartite game; Strategic choices

1. Introduction

In the current economic and social development, the integration of industry and education has become a key measure to promote the deep connection between the education system and the industry to promote the optimal allocation of human resources (Liu, 2022; Zhang & Li, 2023). With the continuous adjustment and upgrading of the industrial structure, a society’s demand for high-quality applied talents is becoming increasingly urgent as this could promote the need for closer cooperation among universities, enterprises, and the government for talent training and industrial development (Chen et al., 2021). In the actual process of promoting the integration of industry and education, the three parties present a complex game phenomenon in decision making based on their respective goals and interests (Bi & Wang, 2018; Nazir & Takeda, 2008). Universities pay attention to the balance of teaching and scientific research, aiming to improve the quality of personnel training, enhance the transformation of scientific research achievements, and improve the reputation of the school through cooperation with enterprises. However, there is a perfunctory phenomenon in the use of practical resources and employment opportunities provided by enterprises, as well as the financial subsidies given by the government (Zeng et al, 2024). Enterprises focus on short-term economic returns and pay more attention to whether they could obtain professional talents in line with their own needs: Reducing costs of human resources and improving technological innovation capabilities through the integration of production and education. Simultaneously, they also face risks such as cost investment, mismatch between talent training and requirements of the enterprises, so there are positive and negative choices in the degree of participation, while the government pursues the maximization of social benefits. The core goal is to promote regional economic development and realize the benign interaction between talent training and the

economic and social development. However, there may be deficiencies in supervision and resource allocation. Due to differences in the goals and interests of the three parties, in the process of carrying out the integration of production and education projects, there are often problems such as insufficient depth of cooperation between universities and enterprises, insufficient motivation for enterprises to participate, and effects of unsatisfactory government policy guidance and supervision, thus seriously affecting the quality and efficiency of the integration of production and education (Howe et al., 2014; Tembrevilla et al., 2024). Therefore, an in-depth study of the evolutionary game relationship of interests among universities, enterprises, and governments in the integration of production and education, in addition to clarifying the interaction mechanism between different decisions of all parties, is helpful to strike a balance and focus of promoting the cooperation of the three parties. This could realize the satisfactory integration of production and education among universities, enterprises, and governments, improve the overall effectiveness of such an integration (Zhou & Etzkowitz, 2021) promote the benign interaction between these two dimensions.

2. Status of the Research

The current research has made remarkable progress in the field of industry-education integration, mainly the reform of integrating industry and education as well as collaborative governance. By constructing the game model and evaluation system (Liu et al., 2023b), it has solved the problems of industry-education integration reform, evaluation dilemma, and collaborative governance. It has been pointed out that there are many problems in the substantive reform of industry-education integration in China, such as the participation of multiple subjects in the “real” and “virtual” of interest integration. These problems make it difficult to “materialize” in “public welfare” in a substantive reform. The fundamental reason is that the foundation of institutional innovation is not solid, rendering it difficult to “soften” the substantive reform (Wang & Wang, 2025).

In accelerating the reform of modern education system, strengthening the integration mechanism of industry and education is the key to promoting the high-quality development of education. By constructing a multi-coordinated operation mechanism, an evaluation mechanism for the integration of industry and education, and an endogenous dynamic development mechanism, the deep integration of production and education elements could be effectively promoted. With the integration of the three chains constituted by education, talent, and industry, a win-win situation could be achieved for multiple subjects in the process of integrating industry and education (Zhang, 2019; Zhuang & Zhou, 2023). However, there are complex behaviors in the integration of industry and education, such as the coexistence of utilitarianism and public welfare. For example, the implementation of the main responsibility of enterprises faces problems such as wasting of resources, insufficient coordination, and unclear responsibilities. It is urgent to clarify the rights and responsibilities of enterprises, promote coordination, and improve laws and regulations to activate motivation of participation (Xu & Yang, 2023). Moreover, most of the existing research focused on the “two-party” game between universities and enterprises or regarded the government as an external policy maker and environmental imposer (Li et al., 2022), but less as an endogenous game subject with its own interest demands and strategic choices (Marens, 2008). This perspective simplifies the complexity of the industry-education integration ecosystem to a certain extent. It is difficult to fully reveal the dynamic evolution mechanism under the intertwined interests of the three parties (Jensen & Tragardh, 2004). Overall, the practical dilemma of the integration of industry and education between schools and enterprises is mainly due to the profit-seeking nature of enterprises, the imperfect training mechanism of universities, and the lack of government guidance (Liu, 2025). These factors lead to the imbalance of interests of the three parties being discussed and the prominent problem of optimizing the interests of industry-education integration. Currently, the research mainly focused on the exploration of the integration mode of industry and education, the construction of collaborative innovation mechanism, and the improvement of policy support system. The analysis of the game process of the government, universities, and enterprises in the integration of industry and education from the perspective of interest evolution is still insufficient (Liu et al., 2023a). Thus, systematic and in-depth theoretical results have not yet been formed. The theory of how to accurately coordinate the interests of the three parties to improve the quality of the integration of industry and education needs to be further explored.

The theoretical contributions of this study are as follows: First, the government is transformed from an external environmental factor into an endogenous independent game player with regular supervision and negative supervision strategies, from which a “university-enterprise-government” tripartite evolutionary game model closer to reality is constructed (Zhang, 2024). Secondly, through this model, we cannot only analyze the cooperative behavior between universities and enterprises, but also reveal how the government’s regulatory strategy affects and is counterproductive by the other two parties’ strategic choices, so as to provide a new dynamic interpretation framework for understanding the dilemma of collaborative governance, such as the integration of industry and education. This bridges the research gap for the lack of discussion on the role of the government as a strategic participant in the existing research and deepens the understanding of the complex governance mechanism of the integration of industry and education.

3. Model Construction

The governance activities of the integration of industry and education involve the dynamic communication of multiple parties. The parties will weigh their own interests in real time according to the dynamic information to decide the kind of governance behavior, thus forming a dynamic evolutionary game process (Steiber & Alänge, 2013). The governance activities of the integration of industry and education involving three parties: universities, enterprises, and governments. Universities release talent training plans in the integration of industry and education. Enterprises participate in the talent training process according to their own needs and the training direction of universities. The government directly supervises the development of industry-education integration (Shou & Li, 2025). Universities achieve talent training through the integration of educational resources; there are two behavioral choices in actively fulfilling educational responsibilities: “Active cooperation” and “negative cooperation”. Enterprises participate in talent cultivation with the help of the integration platform of industry and education. There are two behavioral choices for the quality of talent cultivation and resource input: “Deep participation” and “formal participation”. There are two behavioral choices for the government to supervise universities and enterprises in the integration of industry and education: “Regular supervision” and “negative supervision”.

3.1 Basic Assumptions and Key Parameters

3.1.1 Basic assumptions

Hypothesis 1: Strategy setting

There are two strategic choices of “positive cooperation” and “negative cooperation” in the integration of industry and education in universities. “Positive cooperation” means universities actively connect with enterprises in depth and invest substantial resources and energy in personnel training, curriculum development, scientific research cooperation, and other aspects. “Negative cooperation” is reflected in the fact that universities only maintain a superficial cooperative relationship, where there is only low enthusiasm and initiative to participate in the integration of industry and education. The strategy setting of enterprises involves “deep participation” and “formal participation”. In the process of “deep participation”, enterprises participate in the process of talent training in universities in an all-round way, via investing funds in the construction of practice and training bases as well as providing professional and technical guidance, etc. Under “formal participation”, the enterprise is only participating symbolically. The degree of participation is shallow and the investment resources are limited. The government can choose “regular supervision” or “negative supervision”. “Regular supervision” requires the government to formulate and improve policies, invest regulatory resources, and ensure the orderly development of industry-education integration projects. “Negative supervision” is manifested in the insufficient supervision of the government on the integration of industry and education projects and the inadequate implementation of policies.

Hypothesis 2: Profit and cost setting

When universities actively cooperate, they need to bear the teacher training fees and other costs $C1$, enhance the reputation of the school through the integration of high-quality production and education, obtain the word-of-mouth income S , and be able to receive government subsidies G according to the proportion of $k1$. When universities cooperate negatively, the cost is $C1'$ ($C1' < C1$) and the income is $R1$ ($R1 < S$). Negative cooperation will lead to damage to the reputation of the school and the loss is S' . Although the government can subsidize $k1G$ and use it all for school construction, it will be punished with a penalty fee $F1$ once it faces government inspection.

During the integration of production and education, enterprises deeply participate and invest $C2$. Through the precise matching of talents, they obtain talent matching income L at the same time, according to the proportion of $k2$ to obtain government subsidies. If the enterprise chooses to participate in this form, the capital investment is $C2'$ ($C2' < C2$), the income is $R2$ ($R2 < L$), and it needs to bear the cost of talent training V . The government subsidy $k2G$ obtained by the enterprise under formal participation can all be used for itself, but it will face the penalty cost $F2$ upon the inspection of the government.

Regular supervision of the government requires the investment of funds like subsidies and other supervisory charges $C3$, by promoting the integration of industry and education to achieve social benefits, such as increasing employment rate, promoting industrial upgrading, etc., $N1$, also to obtain invisible benefits like demonstration effect, accumulation database, etc., $N2$. When the government is negatively regulated, the capital is invested in $C3'$ ($C3' < C3$), but it will cause public doubts due to poor supervision and loss of credibility A . Therefore, the income is $N2'$ ($N2' < N2$).

Hypothesis 3: Strategy selection probability setting

The probability of universities choosing positive cooperation is $P = X$, and the probability of choosing negative cooperation is $P = 1 - X$. The probability of enterprises choosing deep participation is Y , and the probability of choosing form participation is $1 - Y$; the probability that the government chooses regular supervision is Z , and the probability of choosing negative supervision is $1 - Z$, where $X, Y, Z \in [0, 1]$. This probability setting reflects the possibility that the three parties choose different strategies in different situations. Through the analysis of

probability changes, we can further explore the dynamic game process and influencing factors of the three parties in the integration of industry and education.

3.1.2 Economic implications of key parameters

To enhance the realistic explanatory power of the model, the economic meaning and practical basis of some key parameters are supplemented as follows:

Government subsidy ($G, k1, k2$). The total subsidy G represents the special financial budget set up by the government to promote the integration of industry and education, such as special funds for education, science, and technology innovation funds, etc. The subsidy ratio $k1$ and $k2$ reflect the policy orientation. In practice, the government may set a higher subsidy ratio (i.e., $k2 > k1$) for deeply involved enterprises according to the degree of urgent need for industrial development to encourage enterprises to invest. This differentiated subsidy is a common means for the government to regulate resource allocation and guide market behavior.

Punishment costs $F1$ and $F2$ not only represent direct economic fines, but also a comprehensive disciplinary measure. In practice, penalties can include recovering government subsidies that have been issued, canceling qualifications for future applications for industry-education integration projects, conducting negative notifications on the official platform, and recording credit files of relevant subjects. The deterrent effect of these measures far exceeds the fine itself, as they constitute a significant opportunity cost for the players to choose negative strategies.

In respect of reputation and credibility (S, A), the loss of university reputation S and the loss of government credibility A are important indirect economic costs in the model. The damage of a university's reputation will directly affect the quality of enrollment, social donations, and opportunities of scientific research cooperation. The decline in the credibility of the government will weaken the efficiency of its future policy implementation and increase the cost of social governance. The quantification of these soft costs into the model enables the game analysis to be more in line with the characteristics of the comprehensive consideration and decision making of each subject in reality.

3.2 Establishment of the Model

In the process of cooperation among universities, enterprises, and governments, their choices of behavior strategies is restricted by many factors. To analyze the influencing factors of the behavior strategies of each participant, the evolutionary game theory was used to construct the evolutionary game model. Firstly, according to the hypothesis, the incomes of universities, enterprises, and the government under different behavior strategies were listed, and then their average expected income was calculated. Finally, the replication dynamic equation was obtained. The tripartite participants of universities, enterprises, and governments obtained incomes according to the income payment matrix in Table 1.

To facilitate understanding, the first item of the table was explained as follows. Universities choose active cooperation, enterprises choose deep participation, and the government chooses regular supervision: The income of universities is the word-of-mouth income S minus the cost of active cooperation $C1$, plus the government subsidy $k1G$ obtained according to the proportion of $k1$, that is $(S - C1 + k1G)$, where other principles are the same.

The income payment matrix and definitions of parameters are illustrated in Table 1 and Table 2.

Table 1. Payoff matrix

Game Entity			Government	
			Regular Supervision (Z)	Passive Regulation (1 - Z)
University	Active cooperation (X)	Deep participation (Y)	$(S + k1G - C1,$ $L + k2G - C2,$ $N1 + N2 - C3)$	$(S + k1G - C1,$ $L + k2G - C2,$ $N2' - C3' - A)$
			$(S + k1G - C1,$ $R2 + k2G - C2' - V - F2,$ $N1 + N2 - C3)$	$(S + k1G - C1,$ $R2 + k2G - C2' - V - F2, N2' -$ $C3' - A)$
	Passive cooperation (1 - X)	Formal participation (1 - Y)	$(R1 + k1G - S' - F1,$ $L + k2G - C2,$ $N1 + N2 - C3)$	$(R1 + k1G - S' - F1,$ $L + k2G - C2,$ $N2' - C3' - A)$
			$(R1 + k1G - S' - F1,$ $R2 + k2G - C2' - V - F2,$ $N1 + N2 - C3)$	$(R1 + k1G - S' - F1,$ $R2 + k2G - C2' - V - F2,$ $N2' - C3' - A)$
Enterprise	Active cooperation (X)	Deep participation (Y)	$(S + k1G - C1,$ $L + k2G - C2,$ $N1 + N2 - C3)$	$(S + k1G - C1,$ $L + k2G - C2,$ $N2' - C3' - A)$
			$(S + k1G - C1,$ $R2 + k2G - C2' - V - F2,$ $N1 + N2 - C3)$	$(S + k1G - C1,$ $R2 + k2G - C2' - V - F2, N2' -$ $C3' - A)$
Enterprise	Passive cooperation (1 - X)	Formal participation (1 - Y)	$(R1 + k1G - S' - F1,$ $R2 + k2G - C2' - V - F2,$ $N1 + N2 - C3)$	$(R1 + k1G - S' - F1,$ $R2 + k2G - C2' - V - F2,$ $N2' - C3' - A)$
			$(S + k1G - C1,$ $L + k2G - C2,$ $N1 + N2 - C3)$	$(S + k1G - C1,$ $L + k2G - C2,$ $N2' - C3' - A)$

Table 2. Definitions of parameters

Parameter	Meaning
$C1$	Costs borne by universities for active cooperation, such as teacher training fees
S	Benefits of reputation gained by universities through high-quality industry-education integration
$k1$	Proportion of government subsidies received by universities
$C1'$	Costs for universities during passive cooperation
$R1$	Benefits for universities during passive cooperation
S'	Damage to universities' reputation due to passive cooperation
$F1$	Penalty fees imposed on universities for passive cooperation during government inspections
$C2$	Funds invested by enterprises for deep participation in industry-education integration
L	Benefits of talent suitability gained by enterprises through precise talent matching
$k2$	Proportion of government subsidies (G) received by enterprises
$C2'$	Funds invested by enterprises during formal participation
$R2$	Benefits for enterprises during formal participation
V	Talent training costs borne by enterprises during formal participation
$F2$	Penalty fees imposed on enterprises for formal participation during government inspections
$C3$	Funds invested by the government for regular supervision
$N1$	Social benefits achieved by the government through promoting industry-education integration
$N2$	Intangible benefits gained by the government through regular supervision
$C3'$	Funds invested by the government during passive supervision
A	Loss of public credibility by the government due to passive supervision
$N2'$	Benefits for the government during passive supervision

Based on the payoff matrix of the three entities, i.e., universities, enterprises, and the government, their expected benefits and average expected benefits under different strategies can be obtained.

4. Establishment of Dynamic Equation

4.1 Strategy Selection Setting of Each Subject

4.1.1 Universities

According to evolutionary game theory, the growth rate of a strategy depends on the difference between its expected return and the average expected return of the group. Define the expected return of the “positive cooperation” strategy as $E11$, the expected return of the “negative cooperation” strategy as $E12$, and the average expected return as $E1$. The replication dynamic equation of the “positive cooperation (X)” strategy is:

$$F(X) = \frac{dX}{dt} = X(E11 - E1)$$

Here, $E1 = X * E11 + (1 - X) E12$. Substituting the average expected return $E1$ into the replication dynamic equation, we obtained:

$$\begin{aligned} F(X) &= X * (E11 - [X * E11 + (1 - X) * E12]) \\ &= X * [(1 - X) E11 - (1 - X) E12] \\ &= X * (1 - X) * (E11 - E12) \end{aligned}$$

Under this model setting, the income of the strategy adopted by the university does not change with the change of the enterprise and the government strategy. $E11$ and $E12$ are calculated according to the income payment matrix.

$$E11 = YZ * (S + k1G - C1) + Y(1 - Z) * (S + k1G - C1) + (1 - Y)Z * (S + k1G - C1) + (1 - Y)(1 - Z) * (S + k1G - C1) = S + k1G - C1;$$

$$E12 = YZ * (R1 + k1G - S' - F1) + Y(1 - Z) * R1 + k1G - S' - F1 + (1 - Y)Z * (R1 + k1G - S' - F1) + (1 - Y)(1 - Z) * (R1 + k1G - S' - F1) = R1 + k1G - S' - F1;$$

$$E1 = X * E11 + (1 - X) * E12;$$

Substituting $E11$ and $E12$ into $F(X) = X * (1 - X) * (E11 - E12)$, we obtained:

$$F(X) = \frac{dX}{dt} = X(1 - X) * [(S + k1G - C1) - (R1 + k1G - S' - F1)]$$

$$F(X) = X(1 - X) * [S - C1 - R1 + S' + F1] \quad (1)$$

Eq. (1) is the replication dynamic equation of university strategy selection. It clearly shows that the direction of evolution and the speed of universities' choice of "positive cooperation" strategy depend on the difference between the net income of "positive cooperation" ($S - C1$) and the net income of "negative cooperation" ($R1 - S' - F1$).

4.1.2 Enterprises

Let the expected return of deep participation in the enterprise be $E21$, the expected return of formal participation is $E22$, and the average expected return is $E2$. Similarly, the replication dynamic equation of the enterprise's choice of "deep participation" (Y) strategy is:

$$F(Y) = \frac{dY}{dt} = Y(E21 - E2) = Y(1 - Y)(E21 - E22)$$

Calculated according to the income payment matrix,

$$E21 = XZ * (L + k2G - C2) + X(1 - Z) * (L + k2G - C2) + (1 - X)Z * (L + k2G - C2) + (1 - X)(1 - Z) * (L + k2G - C2) = L + k2G - C2$$

$$E12 = XZ * (R2 + k2G - C2' - V - F2) + X(1 - Z) * (R2 + k2G - C2' - V - F2) + (1 - X)Z * (R2 + k2G - C2' - V - F2) + (1 - X)(1 - Z) * (R2 + k2G - C2' - V - F2) = R2 + k2G - V - F2$$

$$E2 = Y * E21 + (1 - Y) * E22$$

Substituting $E21$ and $E22$ into $F(Y)$, we obtained:

$$F(Y) = \frac{dY}{dt} = Y(1 - Y) * [(L + k2G - C2) - (R2 + k2G - V - F2)]$$

$$F(Y) = Y(1 - Y) * [L - C2 - R2 + V + F2] \quad (2)$$

Eq. (2) is a replication dynamic equation for enterprise strategy selection. It shows that the evolution of enterprises' choice of "deep participation" strategy depends on the difference between the net income of "deep participation" ($L - C2$) and the net income of "formal participation" ($R2 - V - F2$).

4.1.3 Government

Suppose the government chooses the expected return of active supervision as $E31$, the expected return of negative supervision as $E32$, and the average expected return as $E3$. Similarly, the replication dynamic equation of the government's choice of "regular supervision" (Z) strategy is as follows:

$$F(Z) = \frac{dZ}{dt} = Z(E31 - E3) = Z(1 - Z)(E31 - E32)$$

According to the income payment matrix, it is calculated that:

$$E31 = N1 + N2 - C3$$

$$E32 = N2' - C3' - A$$

$$E3 = Z * E31 + (1 - Z) * E32$$

According to the income payment matrix, it is calculated that $E21$ and $E22$ are substituted into $F(Z)$ to obtain:

$$F(Z) = \frac{dZ}{dt} = Z(1 - Z) * [(N1 + N2 - C3) - (N2' - C3' - A)]$$

$$F(Z) = Z(1 - Z) * [N1 + N2 - C3 - N2' + C3' + A] \quad (3)$$

Eq. (3) is the replication dynamic equation of the government's strategy choice. It shows that the evolution of the government's choice of "regular regulation" strategy depends on the difference between the net income of "regular regulation" ($N1 + N2 - C3$) and the net income of "negative regulation" ($N2' - C3' - A$).

4.2 Equilibrium Point Analysis

If the replication dynamic equation of universities, enterprises, and governments and the probability X , Y , and Z of their respective choices of active, deep, and regular regulatory behavior strategies satisfy the following, their behavior strategies tend to be stable.

$$F(X) = \frac{dx}{dt} = X(E11 - E1) = X(1 - X) * [S - C1 - R1 + S' + F1]$$

$$F(Y) = \frac{dy}{dt} = Y(E21 - E2) = Y(1 - Y) * [L - C2 - R2 + V + F2]$$

$$F(Z) = \frac{dz}{dt} = Z(E31 - E3) = Z(1 - Z) * [N1 + N2 - C3 - N2' + C3' + A]$$

$X, Y, Z \in [0, 1]$, so positive and negative depend on $E11, E12, E21, E22, E31$, and $E32$.

4.2.1 University's equilibrium point

When $F(X) = 0$, i.e. $X = 0$ or $X = 1$, it indicates that the university's strategy choice reaches a balanced state. The critical condition is $X = 0/X = 1$. When $E11 > E12$, i.e., $(S - C1 > R1 - S' - F1)$, $\frac{dX}{dt} > 0$, and $X = 1$ is a stable point, it indicates that universities tend to choose active cooperation. When $E11 < E12$, i.e., $(S - C1 < R1 - S' - F1)$, $\frac{dX}{dt} < 0$, universities tend to choose passive cooperation.

4.2.2 Enterprise's equilibrium point

When $F(Y) = 0$, i.e., $Y = 0$ or $Y = 1$, it indicates that the enterprise's strategy choice reaches a balanced state. When $E21 > E22$, i.e., $(L - C2 > R2 - V - F2)$, $\frac{dY}{dt} > 0$, and $Y = 1$ is a stable point, it shows that all enterprises tend to choose deep participation. When $E21 < E22$, i.e., $(L - C2 < R2 - V - F2)$, $\frac{dY}{dt} < 0$, enterprises tend to choose formal participation

4.2.3 Government's equilibrium point

When $F(Z) = 0$, i.e., $Z = 0$ or $Z = 1$, it demonstrates that the government's strategy choice reaches a balanced state. When $E31 > E32$, i.e., $(N1 + N2 - C3 > N2' - A - C3')$, $\frac{dZ}{dt} > 0$, and $Z = 1$ is a stable point, marking that the government tends to choose regular supervision. When $E31 < E32$, i.e., $(N1 + N2 - C3 < N2' - A - C3')$, $\frac{dZ}{dt} < 0$, the government tends to choose passive supervision.

4.3 Stability Analysis

All possible strategy combinations are $2 * 2 * 2 = 8$: $E1(0,0,0)$, $E2(0,0,1)$, $E3(0,1,0)$, $E4(0,1,1)$, $E5(1,0,0)$, $E6(1,0,1)$, $E7(1,1,0)$, and $E8(1,1,1)$.

The Jacobian matrix was constructed and the equilibrium point was substituted to calculate its eigenvalues.

$$J = \begin{bmatrix} \frac{\partial F1}{\partial x} & \frac{\partial F(X)}{\partial Y} & \frac{\partial F1}{\partial Z} \\ \frac{\partial F2}{\partial x} & \frac{\partial F(Y)}{\partial Y} & \frac{\partial F2}{\partial Z} \\ \frac{\partial F3}{\partial x} & \frac{\partial F(Z)}{\partial Y} & \frac{\partial F3}{\partial Z} \end{bmatrix} = \begin{bmatrix} \frac{\partial F(X)}{\partial x} & 0 & 0 \\ 0 & \frac{\partial F(Y)}{\partial y} & 0 \\ 0 & 0 & \frac{\partial F(Z)}{\partial z} \end{bmatrix}$$

$$= \begin{bmatrix} (1 - 2x) A & 0 & 0 \\ 0 & (1 - 2y) B & 0 \\ 0 & 0 & (1 - 2z) C \end{bmatrix}$$

where, $A = S - C1 - R1 + S' + F1$; $B = L - C2 - R2 + V + F2$; $N1 + N2 - C3 - N2' + C3' + A$

The stability of the equilibrium point was judged according to the eigenvalue symbol of the Jacobian matrix. If the real parts of all eigenvalues are negative, the equilibrium point is stable.

For the equilibrium point $E1(0,0,0)$, the eigenvalues $\lambda1 = A, \lambda2 = B, \lambda3 = C$, that is, when $S - C1 - R1 + S' + F1 < 0$ and $L - C2 - R2 + V + F2 < 0$, and $N1 + N2 - C3 - N2' + C3' + A < 0$, $E1(0,0,0)$ is stable. When the net income of universities, enterprises, and governments choosing positive strategies is less than that of negative strategies, the system will be locked in a low-level equilibrium state with no participation.

For the equilibrium point $E2(0,0,1)$, $\lambda1 = A, \lambda2 = B, \lambda3 = -C$, when $S - C1 - R1 + S' + F1 < 0$ and $L - C2 - R2 + V + F2 < 0$ and $N1 + N2 - C3 - N2' + C3' + A > 0$, $E2(0,0,1)$ is stable.

For the equilibrium point $E3(0,1,0)$, $\lambda1 = A, \lambda2 = -B, \lambda3 = -C$, when $S - C1 - R1 + S' + F1 > 0$ and $N1 + N2 - C3 - N2' + C3' + A < 0$, $E3(0,1,0)$ is stable.

For the equilibrium point $E4(0,1,1)$, $\lambda1 = A, \lambda2 = -B, \lambda3 = -C$, when $S - C1 - R1 + S' + F1 > 0$ and $N1 + N2 - C3 - N2' + C3' + A > 0$, $E4(0,1,1)$ is stable.

For the equilibrium point $E5(1,0,0)$, $\lambda1 = -A, \lambda2 = B, \lambda3 = C$, when $S - C1 - R1 + S' + F1 > 0$ and $L - C2 - R2 + V + F2 < 0$ and $N1 + N2 - C3 - N2' + C3' + A < 0$, then $E5(1,0,0)$ is stable.

For the equilibrium point $E6(1,0,1)$, $\lambda1 = -A, \lambda2 = B, \lambda3 = -C$, when $S - C1 - R1 + S' + F1 > 0$ and $L - C2 - R2 + V + F2 > 0$, then $E6(1,0,1)$ is stable.

For the equilibrium point $E7(1,1,0)$, the eigenvalues $\lambda1 = -A, \lambda2 = -B, \lambda3 = -C$, when $S - C1 - R1 + S' + F1 > 0$ and $L - C2 - R2 + V + F2 > 0$ and $N1 + N2 - C3 - N2' + C3' + A < 0$, then $E7(1,1,0)$ is stable.

For the equilibrium point $E8(1,1,1)$, the eigenvalues $\lambda1 = -A, \lambda2 = -B, \lambda3 = -C$, when $S - C1 - R1 + S' + F1 > 0$ and $L - C2 - R2 + V + F2 > 0$ and $N1 + N2 - C3 - N2' + C3' + A > 0$, then $E8(1,1,1)$ is stable. When the net income of the three parties is greater than 0, the system reaches the ideal Pareto optimal state and the three parties

cooperate and a win-win situation is attained.

According to the above analysis, the evolution result of the system depends on the symbol combination of parameters A , B , and C . With different parameters, the system may converge to different equilibrium points. That is, in the case of insufficient incentive and high cost ($A < 0$, $B < 0$, $C < 0$), the system will converge to $E1(0,0,0)$; in the case of sufficient incentive and controllable cost ($A > 0$, $B > 0$, $C > 0$), the system will converge to the ideal $E8(1,1,1)$.

5. Numerical Simulation and Analysis

To show the dynamic evolution path and final stable state of the tripartite strategy selection of universities, enterprises, and governments under different parameter conditions, this section will use MATLAB software for numerical simulation. According to previous stability analysis, the evolution results of the system are sensitive to the initial values of each parameter. We set the simulation time span as $t \in [0,10]$ and the willingness of the three parties to choose the initial strategy is neutral, so $x = 0.5$, $y = 0.5$, $z = 0.5$.

In order to explore whether the system can evolve to the ideal state (active cooperation, deep participation, and regular supervision), we stipulate a set of parameters to allow the parties higher returns in the ideal state.

5.1 Scenario 1: Insufficient Incentives and Penalties

Firstly, the evolutionary trend of the system was discussed in the absence of effective incentive mechanism and high supervision cost. Suppose that the initial parameters are set as follows:

University parameters: $S = 10$, $C1 = 8$, $R1 = 5$, $S' = 2$, $F1 = 2$;

Enterprise parameters: $L = 12$, $C2 = 10$, $R2 = 6$, $V = 1$, $F2 = 2$;

Government parameters: $N1 = 15$, $N2 = 5$, $C3 = 22$, $N2' = 3$, $C3' = 2$, $A = 4$.

The difference of net income for each party was calculated according to the equilibrium point condition:

The net profit margin of universities: $(S - C1) - (R1 - S' - F1) = (10 - 8) - (5 - 2 - 2) = 2 - 1 = 1 > 0$

The net income difference of enterprises: $(L - C2) - (R2 - V - F2) = (12 - 10) - (6 - 1 - 2) = 2 - 3 = -1 < 0$

Government net income gap: Due to the extremely high regulatory cost ($C = 22$), when the synergistic effect had not yet formed, the net income of government supervision alone was much lower than that of non-supervision (or maintaining the status quo), resulting in insufficient regulatory impetus.

In this scenario, the net income of enterprises choosing “deep participation” was less than that of “formal participation”, so enterprises lacked motivation. The results of evolution are shown in Figure 1.

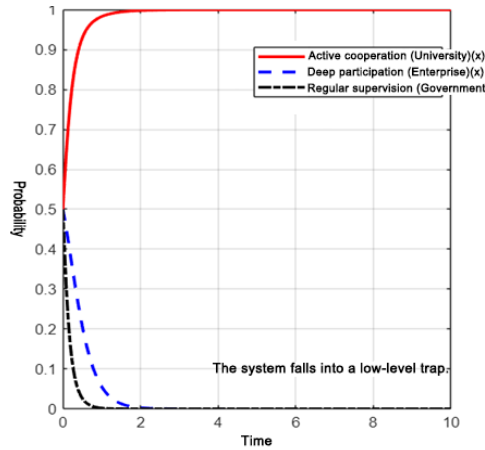


Figure 1. Path of evolution when incentives and punishment are insufficient

The simulation results show that the system quickly falls into the “low level trap”.

Due to the lack of incentives ($L - C2 < R2 - V - F2$), enterprises quickly slid to “formal participation” and the probability y approached 0; with the negative exit of enterprises, universities have lost their partners in collaborative innovation and it is not cost-effective to bear the cost of cooperation alone, so the enthusiasm x will also decrease. Because the regulatory cost C was too high (set to 22), in the case that enterprises and universities were not active, maintaining high-intensity supervision was not only costly but also lacked the support of social benefits. Finally, the government rationally chose to give up supervision, and the probability z tended to be 0. The final system was stable in $(0,0,0)$ state, indicating that if there was no reasonable benefit distribution and cost sharing mechanism, the industry-education integration system would be difficult to maintain and the three parties would choose the “exit” strategy.

5.2 Scenario 2: Scenario Analysis after Strengthening Incentives and Supervision

Based on scenario 1, we adjusted the key parameters to simulate the effect of the government's enhanced policy intervention. The main adjustment strategy is to improve the talent matching income of enterprises (such as through tax incentives), increase the punishment for “formal participation”, and optimize the supervision process to reduce administrative costs. When the enterprise talent matching income was increased to $L = 15$, the punishment was increased to $F = 5$, and the government supervision cost was reduced to $C = 8$. Other parameters remained unchanged.

At that time, the net income gap between the parties had undergone a fundamental reversal:

The net income difference of enterprises: $(L - C_2) - (R_2 - V - F_2) = (15 - 10) - (6 - 1 - 5) = 5 - 0 = 5 > 0$

Government net income gap: With the decrease of C_3 , regulation became “profitable” and the government had the incentive to maintain a high level of regulation.

The results of evolution are shown in Figure 2.

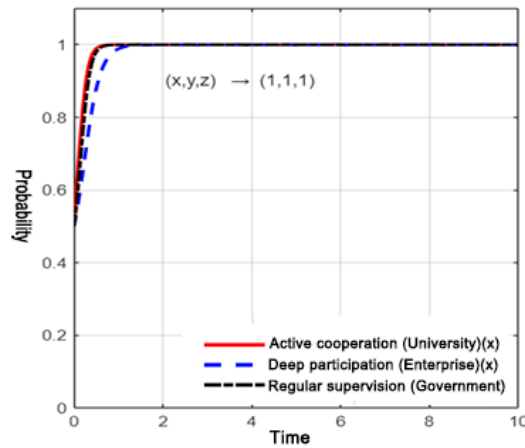


Figure 2. Path of evolution after strengthening incentives and supervision

The simulation results illustrated that when the government's policy tools were enough to change the income structure of enterprises, the system's evolution path had undergone a qualitative leap. Under the dual drive of “high income + high punishment”, the expected income of “deep participation” of enterprises is significantly higher than that of “formal participation” and the probability y rises rapidly and stabilizes at 1. The active participation of enterprises has led to the realization of collaborative benefits. Universities have benefited from cooperation and strategy x tends to be stable at 1; since the government reduces the regulatory cost and improves the overall efficiency of the system, the benefit of maintaining the regulatory strategy is greater than the cost, and the probability z also tends to be 1. Finally, the system successfully evolves to the ideal stable point $E_8(1,1,1)$. This reveals that for the weakest link in the game for enterprises in this particular case, the incentives and constraints of precise policy intervention, when combined with reasonable regulatory cost control, guide the system from “low-level trap” to “Pareto optimality”.

6. Conclusions and Recommendations

6.1 Conclusions

This study employed evolutionary game theory to construct a tripartite evolutionary game model of universities, enterprises, and governments in the integration of industry and education, in order to systematically analyze its stability. The results showed that:

6.1.1 Stability of university's choice of strategy

The stability of universities' choice of “positive cooperation” strategy depended on the difference between the net income of positive cooperation and the net income of negative cooperation. When the net benefit of positive cooperation ($S - C_1$) was greater than the net benefit of negative cooperation ($R_1 - S' - F_1$), that is, $S - C_1 > R_1 - S' - F_1$, universities tended to choose the “positive cooperation” strategy and the strategy was in a stable state. On the contrary, if $S - C_1 < R_1 - S' - F_1$, universities tended to choose the “negative cooperation” strategy and the “negative cooperation” strategy was in a stable state.

6.1.2 Stability of enterprises' strategy selection

The stability of the enterprises' choice of “deep participation” strategy depended on the difference between the net incomes of deep participation and formal participation. When the net income of deep participation ($L - C_2$)

was greater than the net income of formal participation ($R2 - V - F2$), that is, $L - C2 > R2 - V - F2$, enterprises tended to choose the “deep participation” strategy and the strategy was in a stable state. On the contrary, if $L - C2 < R2 - V - F2$, enterprises tended to choose the “formal participation” strategy and the “formal participation” strategy was in a stable state.

6.1.3 Stability of the government’s choice of strategy

The stability of the government’s choice of “regular supervision” strategy depended on the difference between the net incomes of its regular supervision and negative supervision. When the net income of regular supervision ($N1 + N2 - C3$) was greater than the net income of negative supervision ($N2' - C3' - A$), that is, $N1 + N2 - C3 > N2' - C3' - A$, the government tended to choose the “regular supervision” strategy and the strategy was in a stable state. On the contrary, if $N1 + N2 - C3 < N2' - C3' - A$, the government tended to choose the “negative regulation” strategy, which was in a stable state.

Through stability analysis, this paper revealed the stable conditions of each subject’s strategy selection in the process of industry-education integration and provided a theoretical basis for understanding and predicting the behavior of each subject in such an integration.

6.2 Recommendations

6.2.1 Optimize the income structure and guide the active cooperation of universities

In order to improve the reputation of universities, the government and enterprise associations established the “gold standard” certification for the integration of production and education to derive excellent achievements. The employment counterpart rate of students, cooperative R&D projects, and other indicators serve as important weight indicators of “double first-class” construction, subject evaluation, and university ranking, so as to transform soft reputation into hard resources. To reduce the cost of universities participating in cooperation, the government could set up a special fund for “teachers entering enterprises”, subsidize teachers’ salary, and travel during the practice of enterprises, or encourage enterprises to provide convenience for visiting teachers through tax reduction and exemption, in order to reduce the direct input cost of universities. The government could establish a blacklist system for industry-education integration projects. To increase the punishment for the negative cooperation of universities, we should not only recover the subsidies of universities, but also publicly notify them and penalize them to declare all types of government education funding projects in the next one to three years.

6.2.2 Enhance motivation of participation to ensure the depth of enterprises’ participation

In terms of policy, in addition to direct financial subsidies, enterprises should pass the certification of industry-education integration. Enterprises should be given priority in tax incentives, government purchase services, credit financing and other aspects, so that they could consider the trained talents core assets of the future. In terms of sharing the cost of enterprises’ participation and promoting the co-construction and sharing of practice and training bases between universities and enterprises in the region, the government should provide a one-time large subsidy for the construction of the base. In addition, enterprises and universities should pay on demand, thus transforming the high fixed cost of a single enterprise into a lower variable cost. To increase the risk of formal participation apart from setting fines, it is necessary to increase its hidden costs. It is stipulated that only enterprises deeply involved in the integration of industry and education are eligible to participate in the formulation of industry standards, or in apprenticeship projects. Therefore, it is clear that “formal participation” leads to the mismatch of talents and the cost of follow-up training should be fully borne by the enterprise and not included in any subsidy projects.

6.2.3 Innovate the regulatory model to achieve effective government supervision

This is to quantify social benefits, establish a dynamic monitoring and social benefit evaluation system for industry-education integration, and link regional employment rate, the speed of industrial technology upgrading, the degree of talent structure optimization and other indicators with the performance appraisal of regulatory authorities. Macro social benefits could be transformed into endogenous power of government departments. Other measures include reducing the cost of government supervision, embracing digital supervision, and developing a unified information platform for the integration of industry and education. Universities and enterprises are required to fill in key data such as cooperation agreements, capital flows, and student internship logs online. By using big data and AI technology to automatically screen abnormal data, the transformation from “regular comprehensive inspection” to “real-time accurate early warning” is realized and the human and material costs of supervision are greatly reduced. According to the enlightenment of numerical simulation, the implementation of dynamic differentiation policy should be dynamic and adaptive. In the early stage of industry-education integration, the government should intervene with strong financial subsidies and penalties, break the old game deadlock, and guide the system to evolve in the direction of tripartite coordination. When active cooperation of the three parties becomes the norm, the government could gradually turn some subsidies into inclusive policies, rely more on

reputation mechanisms and market forces for adjustment, in order to reduce administrative costs and achieve long-term governance.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflict of interest.

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